

- 2 Cholesterol oxidase is an enzyme produced by some species of bacteria. This enzyme is widely used in research, in diagnostic laboratories and in industry.

A student carried out an investigation into cholesterol oxidase produced by a soil-living bacterium. The bacterium secretes the enzyme to break down cholesterol in the external environment.

- (a) State the term used to describe an enzyme that is synthesised in the cell and secreted to the external environment to act outside the cell.

..... [1]

- (b) The student carried out experiments under standardised conditions of temperature and pH to measure the activity of cholesterol oxidase at different concentrations of substrate.

The student:

- converted the results to a unit for enzyme activity, expressed as U cm^{-3}
- plotted a graph to show the effect of substrate concentration on the activity of cholesterol oxidase
- used the graph to find out the maximum rate of reaction (V_{max}) and derive the Michaelis-Menten constant (K_m) of the enzyme
- determined the value of V_{max} as 580 U cm^{-3}
- derived the value of K_m as $0.15 \text{ mmol dm}^{-3}$.

Figure 2.1 shows the axes of the graph that the student used to plot the experimental results.

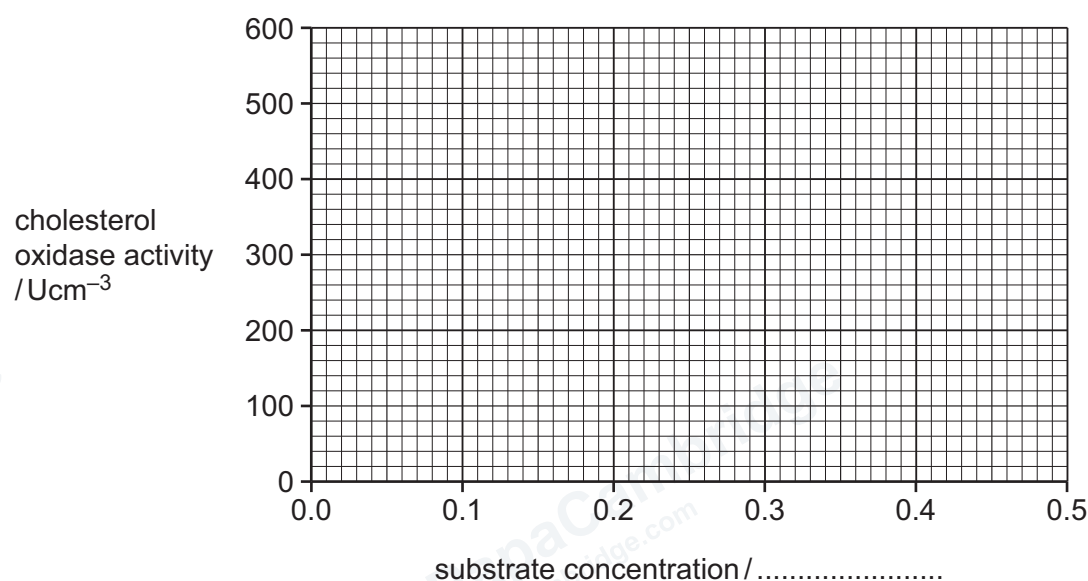


Figure 2.1

- (i) Use the values of V_{max} and K_m obtained by the student to:

- add the correct units for substrate concentration on the dotted line provided in Figure 2.1
- sketch a curve on Figure 2.1 that will provide the same values of V_{max} and K_m as those obtained by the student.

[2]

- (ii) The student then carried out experiments to show the effect of pH on the activity of cholesterol oxidase.

The results are shown in Figure 2.2.

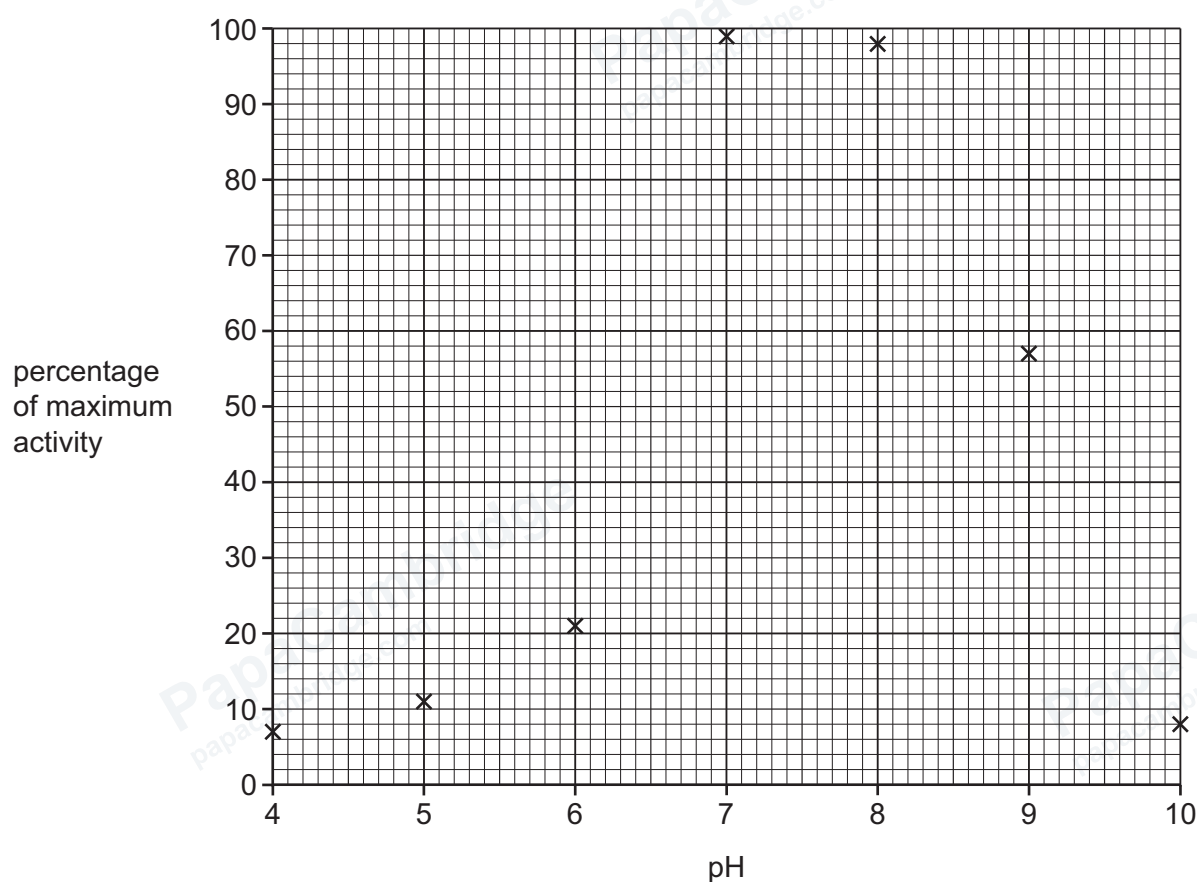


Figure 2.2

The student concluded that the cholesterol oxidase could be useful in the food industry to determine the concentration of cholesterol in food samples.

With reference to Figure 2.1 and Figure 2.2, explain whether you agree with the conclusion made by the student.

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- (c) *Mycobacterium tuberculosis*, the pathogen that causes tuberculosis (TB), synthesises cholesterol oxidase. This enzyme is secreted by the bacterium and is **also** found in the cell surface membrane of the bacterium.

Suggest **and** explain how cholesterol oxidase can help *M. tuberculosis* to enter a host cell and then cause harm to the host cell.

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- (d) There may be benefits in using a chemical that can act on cholesterol oxidase as a drug to treat people with TB and to control the spread of the disease.

Suggest **and** explain the potential benefits in using a chemical that can act on cholesterol oxidase as a drug to treat people with TB and to control the spread of the disease.

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..... [3]

[Total: 12]



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